

Shaurya Pundir

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ACADEMIC DETAILS

Examination	University	Institute	Year	CGPA/%
Graduation	Computer Science and Engineering with Artificial Intelligence	Graphic Era Deemed to be University	2026	7.97/10.0
Intermediate	CBSE	Vivekananda School, DehraDun	2022	78.3
Matriculation	ICSE	The Heritage School, DehraDun	2020	88.3

FIELDS OF INTEREST

- Artificial Intelligence (AI), Large Language Models (LLMs), Data Science
- Neural Networks and Deep Learning
- Computer Vision and related AI-based project development

TECHNICAL SKILLS

- **Languages** (C, C++, Java, Javascript, Python, Pytorch), **Database** (MySQL, MongoDB) **Script** (Python), **Tools** (Eclipse, Power BI, Tableau).

LICENCES & CERTIFICATIONS

- Amazon Web Services(AWS) Certified Cloud Practitioner
- Classification Analysis by University of Colorado Boulder
- Regression Analysis by University of Colorado Boulder
- Introduction to Generative AI by Google
- Microsoft Certified: Azure AI Fundamentals
- Microsoft Certified: Power BI Data Analyst Associate

Projects

- **Machine Learning Based Property Price Prediction** (B.Tech. Project)
(Guide: Mr. Vivek Tomar, July'24)
 - * Objective: Developed a machine learning model to predict house prices using features like square footage, number of bedrooms, and location.
 - * Implemented data preprocessing, feature engineering, and outlier removal on the Bengaluru house price dataset, achieving accurate predictions with Linear Regression and Lasso Regression models.
 - * Evaluated model performance using Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared (R^2), with plans to deploy as a Flask web application.
- **Network Intrusion Detection using Machine Learning** (Mini Project)
(Guide: Dr. Neha Tripathi, Dec'23 – Feb'24)
 - * Objective: Developed a network intrusion detection system using various machine learning algorithms.
 - * Conducted exploratory data analysis and preprocessing on NSL-KDD dataset containing 42 features and over 25,000 records.
 - * Selected top 10 features using Recursive Feature Elimination with Random Forest classifier.
 - * Trained and evaluated KNN, Logistic Regression, and Decision Tree models achieving up to 99.3% test accuracy.

- * Performed hyperparameter tuning using Optuna and validated models via 10-fold cross-validation (precision and recall over 98%).
- * Assessed model performance using confusion matrices, F1-scores, and classification reports.
- **Semantic Clustering of Image Captions from Flickr8k Dataset** (Project)
(Apr'25 – Apr'25)
 - * Objective: Cluster semantically similar image captions to explore unsupervised grouping of human-written descriptions.
 - * Used the Flickr8k dataset containing 8,000+ images and five captions per image for clustering analysis.
 - * Applied TF-IDF and BERT embeddings to convert captions into high-dimensional vectors.
 - * Performed KMeans clustering; evaluated cluster quality using silhouette scores and visualized results using PCA.
 - * Observed significantly improved coherence in clusters using BERT over traditional vectorization techniques.
- **Smart Traffic Monitoring using YOLOv11 and OCR** (Mini Project)
(Guide: Mr. Yuvraj Joshi, Aug'24 – Dec'24)
 - * Objective: Develop a real-time traffic monitoring system for vehicle detection, speed estimation, and license plate recognition.
 - * Fine-tuned YOLOv11 on a custom license plate dataset for object detection; integrated DeepSORT for multi-vehicle tracking.
 - * Estimated vehicle speed using positional tracking across video frames and flagged over-speeding vehicles.
 - * Performed OCR on cropped license plate images using PaddleOCR and logged all results in a MySQL database.
 - * Handled real-time performance through multithreading and frame-skipping optimization; achieved robust tracking across multiple vehicles.
- **Pipeline Scheduling Optimization using CPU Scheduling Algorithms** (PBL Project)
(Guide: [Dr. Deepak Gaur], Feb'25 – April'25)
 - * Objective: Simulate a DevOps pipeline and optimize task scheduling using OS-level CPU scheduling algorithms.
 - * Built CI/CD simulation with stages (Code Commit, Build, Test, Deploy, Monitor), each scheduled via FCFS, SJN, Round Robin, and Priority algorithms.
 - * Deployed backend using Apache Web Server with CGI scripts; containerized via Docker and orchestrated on a Kubernetes cluster (5 pods).
 - * Tracked pipeline metrics (waiting time, CPU usage, throughput) via Python (psutil, time); validated results with automated scripts and kubectl tests.
 - * Designed a front-end dashboard (HTML/CSS/JS, in progress) to visualize turnaround time, throughput, and CPU utilization.

STRENGTHS

- Rapid learner with a hands-on approach to solving problems and can handle work pressure very well until it gets done.
- Strong collaborative mindset with effective cross-functional communication.
- Highly adaptable to dynamic technologies and fast-paced environments.

INTERESTS AND HOBBIES

- Solving logic puzzles and brainteasers to enhance analytical thinking.
- Engaging in running and cycling to maintain mental clarity and physical endurance.
- Listening to podcasts on technology, spirituality, and geopolitics to stay informed and broaden perspectives.